

PRACTICAL QUESTION

Create an Angular application called **shopping-cart** with the following requirements:

1. Create two feature modules:
 - o ProductModule
 - o CartModule
2. Use **Angular Routing** to navigate between pages.
3. Create a navigation bar using **routerLink**.
4. Use **router-outlet** to load pages.
5. Show product list.
6. On clicking a product, open **product details page** using **Route Parameter**.
7. Load both modules using **Lazy Loading**.

WHAT YOU WILL LEARN

- What is **Module**
- What is **Feature Module**
- What is **Routing**
- What is **routerLink**
- What is **router-outlet**
- What is **Route Parameter**
- What is **Lazy Loading**

STEP-BY-STEP PROCEDURE (VERY CLEAR)

STEP 1: Create New Angular Project

👉 Open VS Code

👉 Open Terminal (Ctrl + `)

👉 Type:

```
ng new shopping-cart
```

👉 It will ask:

Would you like to add Angular routing? → YES

Which stylesheet format? → CSS

👉 Now go inside project:

```
cd shopping-cart
```

👉 Run project:

```
ng serve
```

👉 Open browser:

```
http://localhost:4200
```

✅ You will see Angular default page.

STEP 2: Open Project in VS Code

```
code .
```

You will see folders like:

```
src/app
```

STEP 3: Create Feature Modules

? Why?

To divide application into parts (Product part and Cart part)

👉 Type:

```
ng g module product --routing
```

```
ng g module cart --routing
```

Now you will see folders:

```
src/app/product
```

```
src/app/cart
```

Each has:

```
product.module.ts
product-routing.module.ts
```

● STEP 4: Create Components

👉 Type:

```
ng g component product/product-list
ng g component product/product-detail
ng g component cart/cart
```

Now structure:

```
product-list
product-detail
cart
```

● STEP 5: Setup Main Routing (Lazy Loading)

? What is Lazy Loading?

Module loads only when we open that page.

👉 Open:

```
src/app/app-routing.module.ts
```

🔗 Replace code with:

```
import { NgModule } from '@angular/core';
import { RouterModule, Routes } from '@angular/router';

const routes: Routes = [
  { path: '', redirectTo: 'products', pathMatch: 'full' },

  {
    path: 'products',
    loadChildren: () => import('./product/product.module').then(m =>
m.ProductModule)
  },

  {
    path: 'cart',
    loadChildren: () => import('./cart/cart.module').then(m =>
m.CartModule)
  }
];

@NgModule({
  imports: [RouterModule.forRoot(routes)],
  exports: [RouterModule]
})
export class AppRoutingModule { }
```

✅ This is **Lazy Loading**

● STEP 6: Add Menu & Router Outlet

👉 Open:

```
src/app/app.component.html
```

🔗 Delete everything and write:

```
<h2>Shopping Cart Application</h2>

<a routerLink="/products">Products</a> |
<a routerLink="/cart">Cart</a>
```

<hr>

<router-outlet></router-outlet>

? What is router-outlet?

It shows the page of the route you click.

● STEP 7: Setup Product Module Routing

👉 Open:

src/app/product/product-routing.module.ts

📝 Write:

```
import { NgModule } from '@angular/core';
import { RouterModule, Routes } from '@angular/router';
import { ProductListComponent } from
'./product-list/product-list.component';
import { ProductDetailComponent } from
'./product-detail/product-detail.component';
```

```
const routes: Routes = [
  { path: '', component: ProductListComponent },
  { path: ':id', component: ProductDetailComponent }
];
```

```
@NgModule({
  imports: [RouterModule.forChild(routes)],
  exports: [RouterModule]
})
export class ProductRoutingModule { }
```

? What is :id?

It is **Route Parameter**

● STEP 8: Setup Cart Routing

👉 Open:

src/app/cart/cart-routing.module.ts

📝 Write:

```
import { NgModule } from '@angular/core';
import { RouterModule, Routes } from '@angular/router';
import { CartComponent } from './cart/cart.component';
```

```
const routes: Routes = [
  { path: '', component: CartComponent }
];
```

```
@NgModule({
  imports: [RouterModule.forChild(routes)],
  exports: [RouterModule]
})
export class CartRoutingModule { }
```

● STEP 9: Create Product List Page

👉 Open:

product-list.component.ts

📝 Inside class add:

```
products = [
```

```
{ id: 1, name: 'Laptop' },
{ id: 2, name: 'Mobile' },
{ id: 3, name: 'Headphone' }
];
```

👉 Open:

product-list.component.html

📄 Write:

```
<h3>Product List</h3>
```

```
<ul>
  <li *ngFor="let p of products">
    <a [routerLink]="[p.id]">{{ p.name }}</a>
  </li>
</ul>
```

? What is this?

When you click product name, it goes to **detail page**

● STEP 10: Product Detail Page (Route Parameter)

👉 Open:

product-detail.component.ts

📄 Write:

```
import { ActivatedRoute } from '@angular/router';

export class ProductDetailComponent {

  productId: any;

  constructor(private route: ActivatedRoute) {}

  ngOnInit() {
    this.productId = this.route.snapshot.paramMap.get('id');
  }
}
```

👉 Open:

product-detail.component.html

📄 Write:

```
<h3>Product Details</h3>

<p>Product ID: {{ productId }}</p>

<a routerLink="/products">Back to Product List</a>
```

● STEP 11: Cart Page

👉 Open:

cart.component.html

📄 Write:

```
<h3>My Cart</h3>
<p>No items in cart</p>
```

✅ FINAL OUTPUT (WHAT YOU WILL SEE)

1. App opens → Product List
2. Click **Cart** → Cart page opens

3. Click any product → Detail page opens
4. URL will show:
`/products/1`
`/products/2`
5. Product & Cart modules load **only when clicked** → Lazy Loading

How to check the output:

● STEP 1: Open Project Folder

Open **VS Code** and open your project folder:
`shopping-cart`

● STEP 2: Open Terminal in VS Code

Press:
`Ctrl + ``
(or menu: Terminal → New Terminal)

● STEP 3: Start Angular Server

In terminal, type:
`ng serve`
Wait until you see:
`Compiled successfully`

● STEP 4: Open in Browser

Open **Chrome / Edge** and type:
`http://localhost:4200`

● WHAT OUTPUT YOU WILL SEE

✓ **First Screen:**
`Shopping Cart Application`

`Products | Cart`

And below it:
`Product List`
`Laptop`
`Mobile`
`Headphone`

● Click "Cart"

You will see:
`My Cart`
`No items in cart`

● Click "Products"

You will see product list again.

● Click "Laptop"

Page will change to:
`Product Details`
`Product ID: 1`
`Back to Product List`
And in address bar you will see:
`http://localhost:4200/products/1`

⚠ **IF OUTPUT IS NOT SHOWING**

 Try this:

```
ng serve --open
```

(It will automatically open browser)